Chapter 9

Other GLMs

The binomial, Gaussian and Poisson GLMs are by far the most commonly used, but there are a number of less popular GLMs that are useful for particular types of data. The gamma and inverse Gaussian are intended for continuous, skewed responses. In some cases, we are interested in modeling both the mean and the dispersion of the response and so we present dual GLMs for this purpose. The quasi-GLM is a model that is useful for nonstandard responses where we are unwilling to specify the distribution but can state the link and variance functions.

9.1 Gamma GLM

The density of the gamma distribution is usually given by:

$$f(y) = \frac{1}{\Gamma(\nu)}\lambda^\nu y^{\nu-1}e^{-\lambda y} \qquad y > 0$$

where ν describes the shape and λ describes the scale of the distribution. However, for the purposes of a GLM, it is convenient to reparameterize by putting $\lambda = \nu/\mu$ to get:

$$f(y) = \frac{1}{\Gamma(\nu)}\left(\frac{\nu}{\mu}\right)^\nu y^{\nu-1}e^{-\left(\frac{y\nu}{\mu}\right)} \qquad y > 0$$

Now $EY = \mu$ and $\text{var}\, Y = \mu^2/\nu = (EY)^2/\nu$. The dispersion parameter is $\phi = \nu^{-1}$. Here we plot a gamma density with three different values of the shape parameter ν (the scale parameter would just have a multiplicative effect) as seen in Figure 9.1:

```
x <- seq(0,8,by=0.1)
plot(x,dgamma(x,0.75), type="l", ylab="", xlab="", ylim=c(0,1.25),
    ↪ xaxs="i", yaxs="i")
plot(x,dgamma(x,1.0), type="l", ylab="", xlab="", ylim=c(0,1.25), xaxs
    ↪ ="i",yaxs="i")
plot(x,dgamma(x,2.0), type="l", ylab="", xlab="", ylim=c(0,1.25), xaxs
    ↪ ="i",yaxs="i")
```

The gamma distribution can arise in various ways. The sum of ν independent and identically distributed exponential random variables with rate λ has a gamma distribution. The χ^2 distribution is a special case of the gamma where $\lambda = 1/2$ and $\nu = df/2$.

The canonical parameter is $-1/\mu$, so the canonical link is $\eta = -1/\mu$. However, we typically remove the minus (which is fine provided we take account of this in any derivations) and just use the inverse link. We also have $b(\theta) = \log(1/\mu) = -\log(-\theta)$

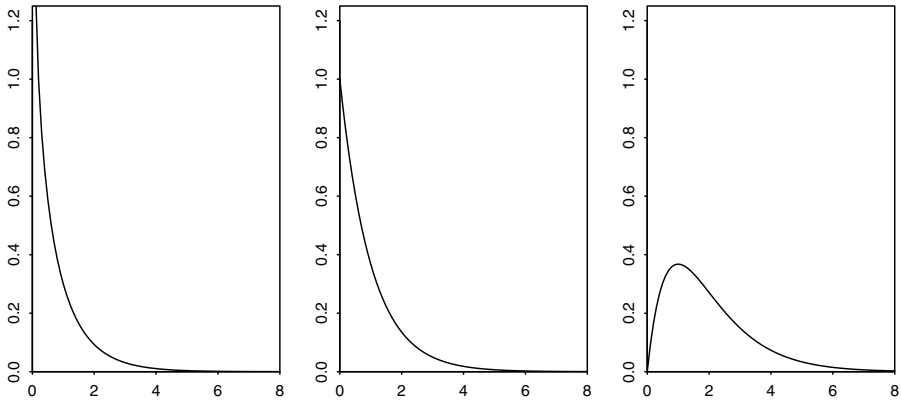


Figure 9.1 *The gamma density explored. In the first panel* $\nu = 0.75$ *and we see that the density is unbounded at zero. In the second panel,* $\nu = 1$ *which is the exponential density. In the third panel,* $\nu = 2$ *and we see a skewed distribution.*

and so $b''(\theta) = \mu^2$ is the variance function. The (unscaled) deviance is:

$$D(y,\hat{\mu}) = -2\sum\{\log y_i/\hat{\mu}_i - (y_i - \hat{\mu}_i)/\hat{\mu}_i\}$$

The utility of the gamma GLM arises in two different ways. Certainly, if we believe the response to have a gamma distribution, the model is clearly applicable. However, the model can also be useful in other situations where we may be willing to speculate on the relationship between the mean and the variance of the response but are not sure about the distribution. Indeed, it is possible to grasp the mean-to-variance relationship from graphical displays with relatively small datasets, while assertions about the response distribution would require a lot more data.

In the Gaussian linear model, var Y is constant as a function of the mean response. This is a fundamental assumption necessary for the optimality of least squares. However, sometimes contextual knowledge of the data or diagnostics shows that var Y is nonconstant. When the form of nonconstancy is known exactly, then weighted least squares can be used. In practice, however, the form is often not known exactly. Alternatively, the transformation of Y may lead to a constant variance model. The difficulty here is that while the original scale Y may be meaningful, $\log Y$ or $\sqrt{Y}$, for example, may not. An example is where a sum of the Ys may be of interest. In such a case, transformation would be a hindrance.

If var $Y \propto EY$, then $\sqrt{Y}$ is the variance stabilizing transform. If one wants to avoid a transformation, a GLM approach can be used. When Y has a Poisson distribution, then var $Y \propto EY$, suggesting the use of a Poisson GLM. Now one might object that the Poisson is a distribution for discrete data, which would seem to disallow its use for continuous responses. However, fitting a GLM only depends on the mean and variance of a distribution; the other moments are not used. This is important because it indicates that we need specify nothing more than the mean and variance. The distribution could be discrete or continuous and it would make no difference.

For some data, we might expect the standard deviation to increase linearly with the response. If so, the coefficient of variation, SD Y/EY, would be constant and var $Y \propto (EY)^2$. For example, measurements of larger objects do tend to have more error than smaller ones.

If we wanted to apply a Gaussian linear model, the log transform is indicated. This would imply a lognormal distribution for the original response. Alternatively, if $Y \sim$ gamma, then var $Y \propto (EY)^2$, so a gamma GLM is also appropriate in this situation. In a few cases, one may have some knowledge concerning the true distribution of the response which would drive the choice. However, in many cases, it would be difficult to distinguish between these two options on the basis of the data alone and the choice would be driven by the purpose and desired interpretation of the model.

There are three common choices of link function:

1. The canonical link is $\eta = \mu^{-1}$. Since $-\infty < \eta < \infty$, the link does not guarantee $\mu > 0$ which could cause problems and might require restrictions on β or on the range of possible predictor values. On the other hand, the reciprocal link has some advantages. The Michaelis–Menten model, used in biochemistry, has:

$$Ey = \mu = \frac{\alpha_0 x}{1 + \alpha_1 x}$$

which can be represented after some reexpression as:

$$\eta = \alpha_1/\alpha_0 + 1/(\alpha_0 x) = \mu^{-1}$$

As x increases, $\eta \to \alpha_1/\alpha_0$, which means that the mean μ will be bounded. The inverse link can be useful in such situations where we know the mean response to be bounded.

2. The log link, $\eta = \log\mu$, should be used when the effect of the predictors is suspected to be multiplicative on the mean. When the variance is small, this approach is similar to a Gaussian model with a logged response.
3. The linear link, $\eta = \mu$, is useful for modeling sums of squares or variance components which are χ^2. This is a special case of the gamma.

The general GLM procedures apply to the analysis and fitting. To estimate the dispersion, McCullagh and Nelder (1989) recommend the use of:

$$\hat{\phi} = \frac{1}{\hat{\nu}} = \frac{X^2}{n-p}$$

The maximum likelihood estimator and the usual estimator, $D/(n-p)$, are both sensitive to unusually small values of the response and are not consistent estimates of the coefficient of variation when the gamma distribution assumption does not hold.

Myers and Montgomery (1997) present data from a step in the manufacturing process for semiconductors. Four factors are believed to influence the resistivity of the wafer and so a full factorial experiment with two levels of each factor was run. Previous experience led to the expectation that resistivity would have a skewed distribution and so the need for transformation was anticipated. We start with a look at the data:

```
data(wafer, package="faraway")
summary(wafer)
```

```
 x1     x2     x3     x4         resist
 -:8    -:8    -:8    -:8    Min.   :166
 +:8    +:8    +:8    +:8    1st Qu.:201
                          Median :214
                          Mean   :229
                          3rd Qu.:259
                          Max.   :340
```

The application of the Box–Cox method or past experience suggests the use of a log transformation on the response. We fit the full model and then reduce it using AIC-based model selection:

```
llmdl <- lm(log(resist) ~ .^2, wafer)
rlmdl <- step(llmdl)
library(faraway)
sumary(rlmdl)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.3111     0.0476  111.53  4.7e-14
x1+           0.2009     0.0476    4.22  0.00292
x2+          -0.2107     0.0476   -4.43  0.00221
x3+           0.4372     0.0673    6.49  0.00019
x4+           0.0354     0.0476    0.74  0.47892
x1+:x3+      -0.1562     0.0673   -2.32  0.04896
x2+:x3+      -0.1782     0.0673   -2.65  0.02941
x3+:x4+      -0.1830     0.0673   -2.72  0.02635

n = 16, p = 8, Residual SE = 0.067, R-Squared = 0.95
```

We find a model with three two-way interactions, all with `x3`.

Now we fit the corresponding gamma GLM and again select the model using the AIC criterion. Note that the family must be specified as `Gamma` rather than `gamma` to avoid confusion with the Γ function. We use the log link to be consistent with the linear model. This must be specified as the default is the inverse link:

```
gmdl <- glm(resist ~ .^2, family=Gamma(link=log), wafer)
rgmdl <- step(gmdl)
sumary(rgmdl)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.3120     0.0476  111.68  4.6e-14
x1+           0.2003     0.0476    4.21  0.00295
x2+          -0.2110     0.0476   -4.44  0.00218
x3+           0.4367     0.0673    6.49  0.00019
x4+           0.0354     0.0476    0.74  0.47836
x1+:x3+      -0.1555     0.0673   -2.31  0.04957
x2+:x3+      -0.1763     0.0673   -2.62  0.03064
x3+:x4+      -0.1819     0.0673   -2.70  0.02687

Dispersion parameter = 0.005
n = 16 p = 8
Deviance = 0.036 Null Deviance = 0.698 (Difference = 0.662)
```

In this case, we see that the coefficients are remarkably similar to the linear model with the logged response. Even the standard errors are almost identical and the square root of the dispersion corresponds to the residual standard error of the linear model:

```
sqrt(summary(rgmdl)$dispersion)
```

```
[1] 0.067267
```

The maximum likelihood estimate of ϕ may be computed using the MASS package:

```
library(MASS)
gamma.dispersion(rgmdl)
```

```
[1] 0.0022657
```

We see that this gives a substantially smaller estimate, which would suggest smaller standard errors. However, it is not consistent with our experience with the Gaussian linear model in this example.

In this example, because the value of $\nu = 1/\phi$ is large (221), the gamma distribution is well approximated by a normal. Similarly, for the logged response linear model, a lognormal distribution with a small variance ($\sigma = 0.0673$) is also very well approximated by a normal. For this reason, there is not much to distinguish these two models. The gamma GLM has the advantage of modeling the response directly while the lognormal has the added convenience of working with a standard linear model.

Let us examine another example where there is more distinction between the two approaches. In Hallin and Ingenbleek (1983) data on payments for insurance claims for various areas of Sweden in 1977 are presented. The data is further subdivided by mileage driven, the bonus from not having made previous claims and the type of car. We have information on the number of insured, measured in policy-years, within each of these groups. Since we expect that the total amount of the claims for a group will be proportionate to the number of insured, it makes sense to treat the log of the number insured as an offset for similar reasons to those in Section 5.3. Attention has been restricted to data from Zone 1. After some model selection, a gamma GLM of the following form was found:

```
data(motorins, package="faraway")
motori <- motorins[motorins$Zone == 1,]
gl <- glm(Payment ~ offset(log(Insured)) + as.numeric(Kilometres) +
    ↪ Make+Bonus, family=Gamma(link=log), motori)
sumary(gl)
```

```
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)             6.5273     0.1777   36.72  < 2e-16
as.numeric(Kilometres)  0.1201     0.0311    3.85  0.00014
Make2                   0.4070     0.1782    2.28  0.02313
Make3                   0.1553     0.1796    0.87  0.38767
Make4                  -0.3439     0.1915   -1.80  0.07355
Make5                   0.1447     0.1810    0.80  0.42473
Make6                  -0.3456     0.1782   -1.94  0.05352
Make7                   0.0614     0.1824    0.34  0.73689
Make8                   0.7504     0.1873    4.01 0.000079
Make9                   0.0320     0.1782    0.18  0.85778
Bonus                  -0.2007     0.0215   -9.33  < 2e-16

Dispersion parameter = 0.556
n = 295 p = 11
Deviance = 155.056 Null Deviance = 238.974 (Difference = 83.918)
```

In comparison, the lognormal model, where we have used the `glm` function for compatibility, looks like this:

```
llg <- glm(log(Payment) ~ offset(log(Insured))+as.numeric(Kilometres)+
    ↪ Make+Bonus,family=gaussian ,  motori)
sumary(llg)
```

```
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)            6.51403    0.18634   34.96  < 2e-16
```

```
as.numeric(Kilometres)  0.05713    0.03265    1.75   0.0813
Make2                   0.36387    0.18686    1.95   0.0525
Make3                   0.00692    0.18824    0.04   0.9707
Make4                  -0.54786    0.20076   -2.73   0.0067
Make5                  -0.02179    0.18972   -0.11   0.9087
Make6                  -0.45881    0.18686   -2.46   0.0147
Make7                  -0.32118    0.19126   -1.68   0.0942
Make8                   0.20958    0.19631    1.07   0.2866
Make9                   0.12545    0.18686    0.67   0.5025
Bonus                  -0.17806    0.02254   -7.90  6.2e-14

Dispersion parameter = 0.611
n = 295 p = 11
Deviance = 173.529 Null Deviance = 238.565 (Difference = 65.035)
```

Notice that there are now important differences between the two models. We see that mileage class given by `Kilometers` is statistically significant in the gamma GLM, but not in the lognormal model. Some of the coefficients are quite different. For example, we see that for make 8, relative to the reference level of make 1, there are $\exp(0.7504) = 2.1178$ times as much payment when using the gamma GLM, while the comparable figure for the lognormal model is $\exp(0.20958) = 1.2332$.

These two models are not nested and have different distributions for the response, which makes direct comparison problematic. The AIC criterion, which is minus twice the maximized likelihood plus twice the number of parameters, has often been used as a way to choose between models. Smaller values are preferred. However, when computing a likelihood, it is common practice to discard parts that are not functions of the parameters. This has no consequence when models with same distribution for the response are compared since the parts discarded will be equal. For responses with different distributions, it is essential that all parts of the likelihood be retained. The large difference in AIC for these two models indicates that this precaution was not taken. Nevertheless, we note that the null deviance for both models is almost the same while the residual deviance is smaller for the gamma GLM. This improvement relative to the null indicates that the gamma GLM should be preferred here. Note that purely numerical comparisons such as this are risky and that some attention to residual diagnostics, scientific context and interpretation is necessary.

We compare the shapes of the distributions for the response using the dispersion estimates from the two models, as seen in Figure 9.2:

```
x <- seq(0,5,by=0.05)
plot(x,dgamma(x,1/0.55597,scale=0.55597),type="l",ylab="", xlab="",
     ↪ yaxs="i",ylim=c(0,1))
plot(x,dlnorm(x,meanlog=-0.30551,sdlog=sqrt(0.61102)),type="l", ylab="
     ↪ ",xlab="",yaxs="i",ylim=c(0,1))
```

We see the greater peakedness of the lognormal indicating more small payments which are balanced by more large payments. The lognormal thus has higher kurtosis.

We may also make predictions from both models. Here is a plausible value of the predictors:

```
x0 <- data.frame(Make="1",Kilometres=1,Bonus=1,Insured=100)
```

and here is predicted response for the gamma GLM:

```
predict(gl,new=x0,se=T,type="response")
$fit
[1] 63061
```

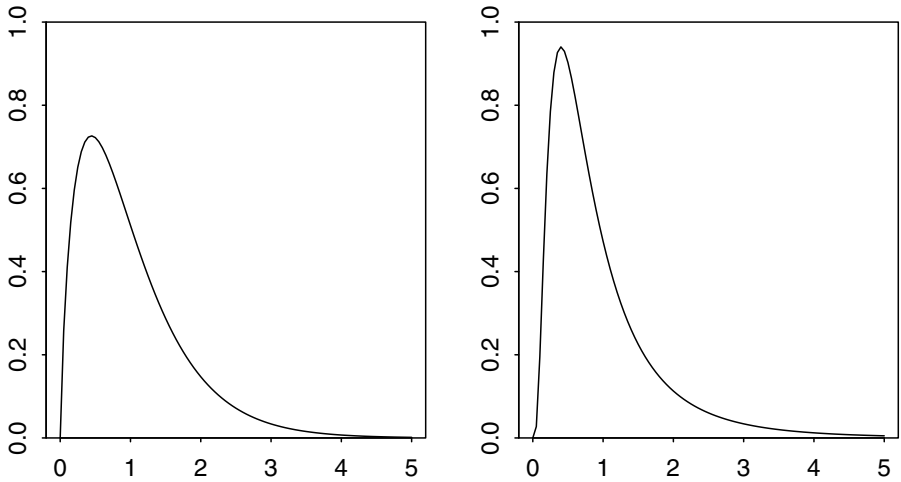


Figure 9.2 *Gamma density for observed shape of 1/0.55597 is shown on the left and lognormal density for an observed SD on the log scale of* $\sqrt{0.61102}$*. The means have been set to one in both cases.*

```
$se.fit
[1] 9711.5
```

For the lognormal, we have:

```
predict(llg,new=x0,se=T,type="response")
$fit
[1] 10.998

$se.fit
[1] 0.16145
```

so that the corresponding values on the original scale would be:

```
c(exp(10.998),exp(10.998)*0.16145)
[1] 59754.5  9647.4
```

where we have used the delta method to estimate the standard error on the original scale.

9.2 Inverse Gaussian GLM

The density of an inverse Gaussian random variable, $Y \sim IG(\mu,\lambda)$, is:

$$f(y|\mu,\lambda) = (\lambda/2\pi y^3)^{1/2}\exp[-\lambda(y-\mu)^2/2\mu^2 y] \qquad y,\mu,\lambda > 0$$

The mean is μ and the variance is μ^3/λ. The canonical link is $\eta = 1/\mu^2$ and the variance function is $V(\mu) = \mu^3$. The deviance is given by:

$$D = \sum_i (y_i - \hat{\mu}_i)^2/(\mu_i^2 y_i)$$

Plots of the inverse Gaussian density for a range of values of the shape parameter, λ, are shown in Figure 9.3. We require the `SuppDists` package of Wheeler (2013).

```
library(SuppDists)
x <- seq(0,8,by=0.1)
plot(x,dinvGauss(x,1,0.5),type="l",ylab="",xlab="",ylim=c(0,1.5),
  xaxs="i",yaxs="i")
plot(x,dinvGauss(x,1,1),type="l",ylab="",xlab="",ylim=c(0,1.5),
  xaxs="i",yaxs="i")
plot(x,dinvGauss(x,1,5),type="l",ylab="",xlab="",ylim=c(0,1.5),
  xaxs="i",yaxs="i")
```

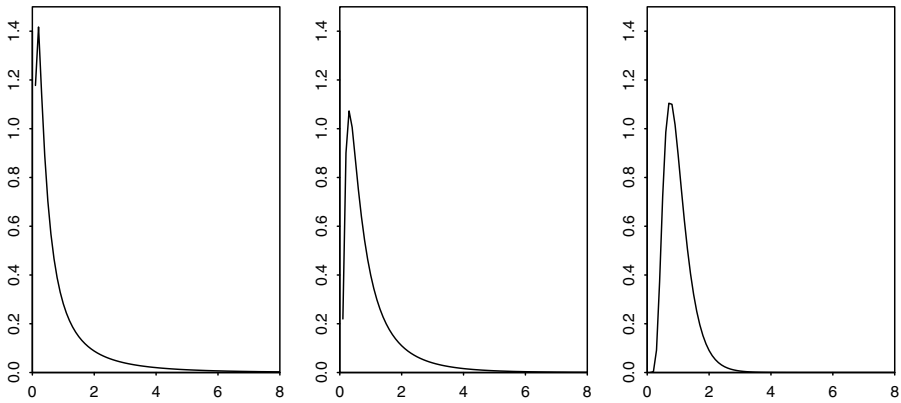


Figure 9.3 *Inverse Gaussian densities for* $\lambda = 0.5$ *on the left,* $\lambda = 1$ *in the middle and* $\lambda = 5$ *on the right.* $\mu = 1$ *in all three cases.*

The case of $\mu = 1$ is known as the *Wald* distribution. The inverse Gaussian has found application in the modeling of lifetime distributions with nonmonotone failure rates and in the first passage times of Brownian motions with drift. See Seshadri (1993) for a book-length treatment.

Notice that the variance function for the inverse Gaussian GLM increases more rapidly with the mean than the gamma GLM, making it suitable for data where this occurs.

In Whitmore (1986), some sales data on a range of products is presented for the projected, x_i, and actual, y_i, sales for $i = 1, \ldots, 20$. We consider a model, $y_i = \beta x_i$ where β would represent the relative bias in the projected sales. Since the sales vary over a wide range from small to large, a normal error would be unreasonable because Y is positive and violations of this constraint could easily occur. We start with a look at the normal model:

```
data(cpd, package="faraway")
lmod <- lm(actual ~ projected-1,cpd)
summary(lmod)
Coefficients:
          Estimate Std. Error t value Pr(>|t|)
projected   0.9940     0.0172    57.9   <2e-16
plot(actual ~ projected, cpd)
abline(lmod)
```

Now consider the inverse Gaussian GLM where we must specify an identity link because we have $y_i = \beta x_i$:

```
igmod <- glm(actual ~ projected-1, family=inverse.gaussian(link="
    ↪ identity"), cpd)
sumary(igmod)
          Estimate Std. Error t value Pr(>|t|)
projected   1.1036     0.0614      18  2.2e-13

Dispersion parameter = 0.000
n = 20 p = 1
Deviance = 0.003 Null Deviance = Inf (Difference = Inf)
abline(igmod,lty=2)
```

We see that there is a clear difference in the estimates of the slope. The fits are shown in the first panel of Figure 9.4. We should check the diagnostics on the inverse

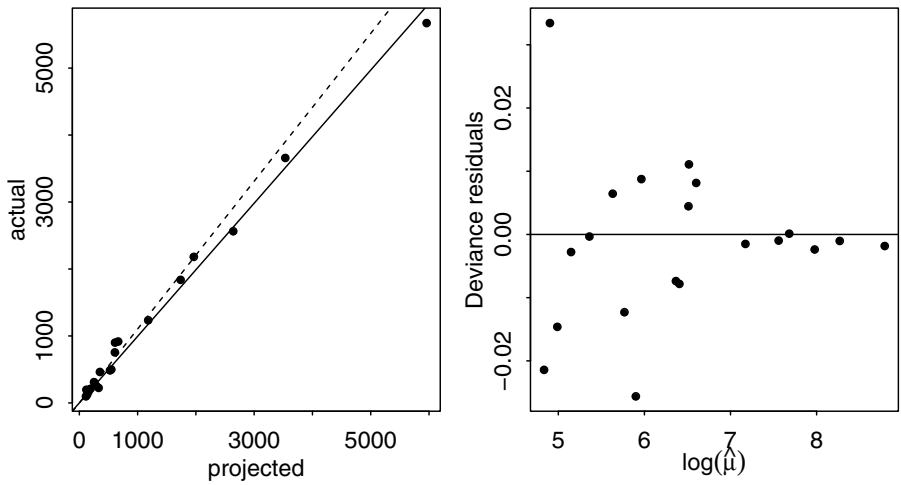


Figure 9.4 *Projected and actual sales are shown for 20 products on the left. The linear model fit is shown as a solid line and the inverse Gaussian GLM fit is shown with a dotted line. A residual-fitted plot for the inverse Gaussian GLM is shown on the right.*

Gaussian GLM:

```
plot(residuals(igmod) ~ log(fitted(igmod)),ylab="Deviance residuals",
    ↪ xlab=expression(log(hat(mu))))
abline(h=0)
```

We see in the second panel of Figure 9.4 that the variance of the residuals is decreasing with error, indicating that the inverse Gaussian variance function is too strong for this data. We have used $\log(\hat{\mu})$ so that the points are more evenly spread horizontally making it easier, in this case, to see the variance relationship. A gamma GLM is a better choice here. In Whitmore (1986), a different variance function is used, but we do not pursue this here as this would not be a GLM.

9.3 Joint Modeling of the Mean and Dispersion

All the models we have considered so far have modeled the mean response $\mu = EY$ where the variance takes a known form: $\text{var}\, Y_i = \phi V(\mu_i)$ where the dispersion parameter ϕ is the variance in the Gaussian model, the squared coefficient of variation in the gamma model and one in the binomial and Poisson models. We can generalize a little to allowing weights by letting $\phi \equiv \phi_i = \phi w_i$ when the weights are known.

In this section, we are interested in examples where ϕ_i varies with the covariates X. This is a particular issue that arises in industrial experiments. We wish to manufacture an item with a target mean or optimized response. We set the predictors to produce items as close as possible to the target mean or to optimize the mean. This requires a model for the mean. We would also prefer that the variance of the response be small at the chosen value of the predictors for production. So we need to model the variance as a function of the predictors.

We take, as an example, an experiment to determine which recipe will most reliably produce the best cake. The data comes from Box et al. (1988) and is shown in Table 9.1. The objective is to bake a cake reliably no matter how incompetent the

Design Vars				Environmental Vars				
			T	0	−	+	−	+
F	S	E	t	0	−	−	+	+
0	0	0		6.7	3.4	5.4	4.1	3.8
−	−	−		3.1	1.1	5.7	6.4	1.3
+	−	−		3.2	3.8	4.9	4.3	2.1
−	+	−		5.3	3.7	5.1	6.7	2.9
+	+	−		4.1	4.5	6.4	5.8	5.2
−	−	+		6.3	4.2	6.8	6.5	3.5
+	−	+		6.1	5.2	6.0	5.9	5.7
−	+	+		3.0	3.1	6.3	6.4	3.0
+	+	+		4.5	3.9	5.5	5.0	5.4

Table 9.1 *F=Flour, S=Shortening, E=Eggs, T=Oven temperature and t=Baking time. "+" indicates a higher-than-normal setting while "−" indicates a lower-than-normal setting. "0" indicates the standard setting.*

cook. For this data, we can see, by examination, that a response of 6.8 is possible for lower flour and shortening content and higher egg content if the temperature is on the high side and the cooking time on the short side. However, we cannot be sure that the consumer will be able to set the temperature and cooking time correctly. Perhaps their oven is not correctly calibrated or they are just incompetent. If they happen to bake the cake for longer than the set time, they will produce a cake with a 3.5 rating. They will blame the product and not themselves and not buy that mix again. If on the other hand we produce the mix with high flour and eggs and low shortening, the worst the customer can do is a 5.2 and will do better than that for other combinations of time and temperature.

Here we need a combination of a high mean with respect to the design factors,

flour, eggs and shortening, and a low variance with respect to the environmental factors, temperature and time. In this example, the answer is easily seen by inspection, but usually more formal model fitting methods will be needed.

Joint Model Specification: We use the standard GLM approach for the mean:

$$EY_i = \mu_i \qquad \eta_i = g(\mu_i) = \sum_j x_{ij}\beta_j \qquad \text{var } Y_i = \phi_i V(\mu_i) \qquad w_i = 1/\phi_i$$

Now the dispersion, ϕ_i, is no longer considered fixed. Suppose we find an estimate, d_i, of the dispersion and model it using a gamma GLM:

$$Ed_i = \phi_i \qquad \zeta_i = \log(\phi_i) = \sum_j z_{ij}\gamma_j \qquad \text{var } d_i = \tau\phi_i^2$$

Notice the connection between the two models. The model for the mean produces the response for the model for the dispersion, which in turn produces the weights for the mean model. In principle, something other than a gamma GLM could be used for the dispersion, although since we wish to model a strictly positive, continuous and typically skewed dispersion, the gamma is the obvious choice. The dispersion predictors Z are usually a subset of the mean model predictors X.

For unreplicated experiments, r_P^2 and r_D^2 are two possible choices for d_i. If replications are available, then a more direct estimate of dispersion would be possible. For more details on the formulation, estimation and inference for these kinds of model, see McCullagh and Nelder (1989), Box and Meyer (1986), Bergman and Hynen (1997) and Nelder et al. (1998).

In the last three citations, data from a welding-strength experiment was analyzed. There were nine two-level factors and 16 unreplicated runs. Previous analyses have differed on which factors are significant for the mean. We found that two factors, `Drying` and `Material`, were apparently strongly significant, while the significance of others, including `Preheating`, was less clear. We fit a linear model for the mean using these three predictors:

```
data(weldstrength, package="faraway")
lmod <- lm(Strength ~ Drying + Material + Preheating, weldstrength)
sumary(lmod)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   43.625      0.262  166.25  < 2e-16
Drying         2.150      0.262    8.19  2.9e-06
Material      -3.100      0.262  -11.81  5.8e-08
Preheating    -0.375      0.262   -1.43     0.18

n = 16, p = 4, Residual SE = 0.525, R-Squared = 0.95
```

Following a suggestion of Smyth et al. (2001), we use the squared studentized residuals, $(y_i - \hat{y}_i)^2/(1-h_i)$, as the response in the dispersion with a gamma GLM using a log-link and weights of $1-h_i$. Again, we follow the suggestion of some previous authors as to which predictors are important for modeling the dispersion:

```
h <- influence(lmod)$hat
d <- residuals(lmod)^2/(1-h)
gmod <- glm(d ~ Material+Preheating,family=Gamma(link=log),
    ↪ weldstrength,weights=1-h)
```

Now feed back the estimated weights to the linear model:

```
w <- 1/fitted(gmod)
lmod <- lm(Strength ~ Drying + Material + Preheating,  weldstrength,
    ↪ weights=w)
```

We now iterate until convergence, where we find that:

```
sumary(lmod)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   43.825      0.108  406.83  < 2e-16
Drying         1.869      0.045   41.53  2.5e-14
Material      -3.234      0.108  -30.03  1.2e-12
Preheating    -0.239      0.101   -2.35    0.036

n = 16, p = 4, Residual SE = 1.000, R-Squared = 1
```

We note that `Preheating` is now significant in contrast to the initial mean model fit. The output for the dispersion model is:

```
sumary(gmod)
            Estimate Std. Error t value  Pr(>|t|)
(Intercept)   -3.064      0.356   -8.60 0.0000010
Material      -3.037      0.413   -7.35 0.0000056
Preheating     2.904      0.413    7.03 0.0000089

overdispersion parameter = 0.500
n = 16 p = 3
Deviance = 20.943 Null Deviance = 57.919 (Difference = 36.976)
```

The standard errors are not correct in this output and further calculation, described in Smyth et al. (2001), would be necessary. This would result in somewhat larger standard errors (about twice the size), but the two factors would still be significant.

9.4 Quasi-Likelihood GLM

An examination of the fitting procedure for GLMs as described in Section 8.2 reveals that it is necessary only to know the link and variance functions to fit the model. This might be convenient since we may wish to minimize the assumptions by avoiding the specification of a distribution for the response. Even so, the distribution is necessary to compute the likelihood and hence the deviance. We need this for the inference. Fortunately, we can compute a substitute for the real likelihood, called the quasi-likelihood. This has already been demonstrated in Sections 3.5 and 5.2 where the quasi-likelihood method is introduced. We return to the idea again later in Section 13.2 in the context of dependent data.

We show here how the quasi method can help clarify the choice of the link and power function. Let's return to the `wafer` example of Section 9.1. We refit the gamma GLM model with just the main effects for simplicity:

```
gl <- glm(resist ~ x1 + x2 + x3 + x4, family=Gamma(link=log), wafer)
```

The equivalent quasi version of this model is:

```
glq <- glm(resist ~ x1 + x2 + x3 + x4, family=quasi(link=log,variance=
    ↪ "mu^2"), wafer)
```

Notice that we need to specify both the link and the variance functions. We compare the deviance, degrees of freedom and log-likelihood for these two models. First the gamma GLM:

```
c(deviance(gl),df.residual(gl),logLik(gl))
[1]   0.12418  11.00000 -70.45392
```

and now the quasi GLM:

```
c(deviance(glq),df.residual(glq),logLik(glq))
```

```
[1]  0.12418 11.00000       NA
```

The fitted coefficients, the deviance and the degrees of freedom are the same for both models. There is no log-likelihood for the quasi GLM because it is not a true likelihood fit. There's not much to recommend the quasi GLM in this instance as the additional assumption of the gamma distribution of the response makes the analysis easier. We can check this assumption so we are not being presumptuous.

The quasi GLM becomes more interesting if we vary the link function. We can consider link functions of the form $\eta = \mu^p$ using the `power` function. We consider values of p in $[0,1]$, where $p = 0$ is considered as $\log\mu$. We fit a model for a range of powers and save the deviance obtained.

```
linkpow <- seq(0,1,by=0.1)
devpow <- numeric(length(linkpow))
for(i in seq(along=linkpow)){
    glqq <- glm(resist ~ x1 + x2 + x3 + x4, family=quasi(link=power(
        ↪ linkpow[i]),variance="mu^2"), wafer)
    devpow[i] <- deviance(glqq)
}
```

A plot of the deviance against the power is shown in the first panel of Figure 9.5. We see that log is indeed the best choice. We would like to investigate $p < 0$ but that facility is not available in the `power` function.

```
plot(linkpow, devpow, type="l", xlab="Link power", ylab="Deviance")
```

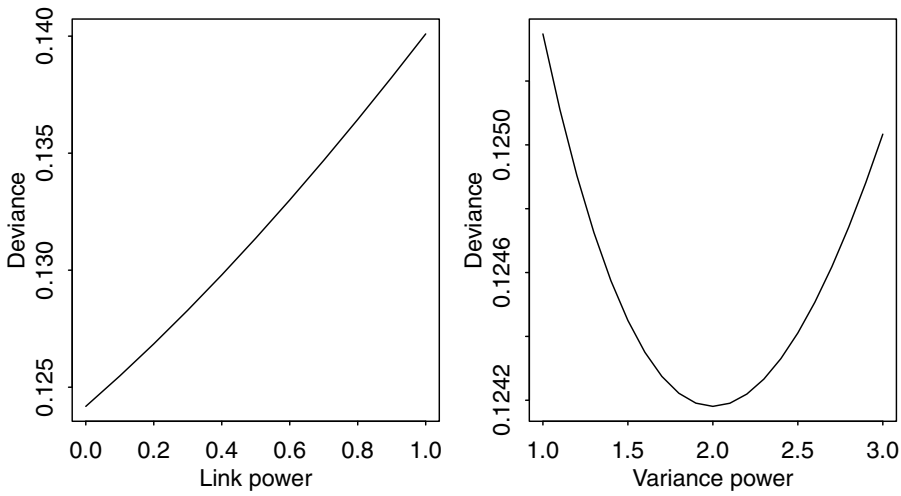


Figure 9.5 *Deviance as the power of the link function is varied on the left and the variance function on the right.*

We can also change the variance function. We might consider variance functions of the form $V(\mu) = \mu^p$. Again we can investigate how the fit of the model changes over a range of values of p, in this case in the range $[1,3]$:

```
powfam <- quasi(link="log",variance="mu^2")
varpow <- seq(1,3,by=0.1)
```

```
devpow <- numeric(length(varpow))
for(i in seq(along=varpow)){
    powfam[["variance"]] <- function(mu) mu^varpow[i]
    glqq <- glm(resist ~ x1 + x2 + x3 + x4, family=powfam, wafer)
    devpow[i] <- deviance(glqq)
}
```

The relationship is shown in the second panel of Figure 9.5. We see that $p = 2$ is the best choice. Fortunately, that choice is implicit in the use of a gamma GLM so we are vindicated.

```
plot(varpow, devpow, type="l", xlab="Variance power", ylab="Deviance")
```

9.5 Tweedie GLM

In Section 8.1, we derived the mean and variance for an exponential (dispersion) family as

$$EY = \mu = b'(\theta) \qquad \text{var}\, Y = b''(\theta)\phi$$

The variance function $V(\mu) = b''(\theta)$ defines how the variance is connected to the mean. We can essentially define the distribution by specifying the variance function since μ can be derived from this. Suppose we consider variance functions of the form $V(\mu) = \mu^p$. We have already seen four examples: the Gaussian ($p = 0$), the Poisson ($p = 1$), the gamma ($p = 2$) and the inverse Gaussian ($p = 3$).

We could take the quasi-likelihood approach and simply choose the link and the variance functions as demonstrated in the previous section. We can specify our chosen link and variance and use the `quasi` family in fitting the `glm()`. Unfortunately, only the four values of $p = 0, 1, 2, 3$ are pre-programmed in base R. For these four choices, it is usually more convenient to use the one of the four distributional GLMs rather than the `quasi` model since we have the advantage of a full rather than quasi-likelihood. We could use other, perhaps noninteger values of p as demonstrated earlier but it is difficult to interpret such choices meaningfully.

Fortunately, any choice of p, except for the open interval $(0, 1)$, defines a so-called Tweedie distribution. Values of p in the interval $(1, 2)$ are particularly interesting since these result in a compound Poisson-Gamma distribution. Observations from this distribution are generated as the sum of a Poisson number of gamma distributed variables in the following manner:

$$Y = \sum_{i=1}^{N} X_i, \quad N \sim \text{Poisson and, independently, } X_i \sim \text{Gamma}$$

The mean of the Poisson is given by $\mu^{2-p}/((2-p)\phi)$. The gamma variables have shape parameter $(2-p)/(p-1)$ and scale parameter $\phi(p-1)\mu^{p-1}$.

The interesting feature of this distribution is the possibility that $N = 0$ and so $Y = 0$. This is useful for modeling responses where there is a nonzero probability that $Y = 0$ but is otherwise a positive value. One example of this is insurance claims by customers that are often zero but can be positive. Rainfall amounts in a short period of time can often be zero but are otherwise some positive quantity. The zero inflated Poisson discussed in Section 5.5 also models excess zeroes but otherwise

the distribution is integer so may not be suitable. Further discussion of the Tweedie distribution may be found in Dunn and Smyth (2005).

The chredlin dataset concerns insurance redlining in Chicago in the 1970s. The response, involact, is the rate of usage of a default insurance plan available to those denied regular insurance in a given zip code. We want to relate this to five demographic predictors. A worked analysis of this data as a standard linear model can be found in Faraway (2014). The response has several zero values out of 47 total observations.

```
data(chredlin, package="faraway")
sum(chredlin$involact == 0)
```

```
[1] 15
```

This makes the assumption of normal distribution for the response somewhat problematic. We could try a Tweedie GLM but we have no particular preference for the value of p. We consider this an additional parameter that can be estimated by likelihood methods. An implementation is found in the mgcv package of Wood (2006):

```
library(mgcv)
twmod <- gam(involact ~ race+fire+theft+age+log(income),  family=tw(
    ↪ link="log"), data=chicago)
summary(twmod)
```

```
Family: Tweedie(p=1.108)
Link function: log
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.69565    9.10630   -1.50  0.14025
race          0.02159    0.00554    3.90  0.00035
fire          0.04850    0.01519    3.19  0.00270
theft        -0.01022    0.00504   -2.03  0.04895
age           0.03458    0.00911    3.80  0.00048
log(income)   1.03099    0.93234    1.11  0.27526

R-sq.(adj) =  0.753   Deviance explained = 69.7%
-REML = 42.572  Scale est. = 0.30847   n = 47
```

The gam function is used instead of glm but otherwise the syntax is familiar. We are free to specify a link function and we make the default choice of log. It seems reasonable that the predictors would have a multiplicative rather than additive effect on the mean response. In the standard linear model, this would be achieved by logging the response, but that is not possible when some response values are zero. We avoid that obstacle by using the log in the link. We see that the estimated value of p is 1.108. Otherwise, the general significance and direction of the effect of the predictors and the quality of the fit are quite similar to that achieved by the linear model.

Consider the first zip code in the dataset. What is the predicted distribution for this case? We set up a grid to compute the density and extract the p and ϕ values:

```
xgrid <- seq(1e-10, 1.25, len=100)
p <- 1.108
phi <- 0.30847
```

The expected mean response for the first case is:

```
(mu <- twmod$fit[1])
```

```
[1] 0.17818
```

From this we can compute the mean of the Poisson as

```
(poismean <- mu^(2-p)/(phi * (2- p)))
```

```
[1] 0.78018
```

which means the probability of a zero response is:

```
(p0 <- exp(-poismean))
```

```
[1] 0.45832
```

Now we compute the density for the nonzero part of the response, scale it down for the probability of a nonzero response and plot. We add a line segment to denote the point mass at zero and show the result in Figure 9.6.

```
twden <- exp(ldTweedie(xgrid, mu, p=p, phi=phi)[,1])
plot(xgrid, twden*(1-p0), type="l",xlab="x",ylab="Density")
dmax <- max(twden*(1-p0))
segments(0,0,0,dmax,lwd=2)
text(0.05,dmax,paste0("p=",signif(p0,3)))
```

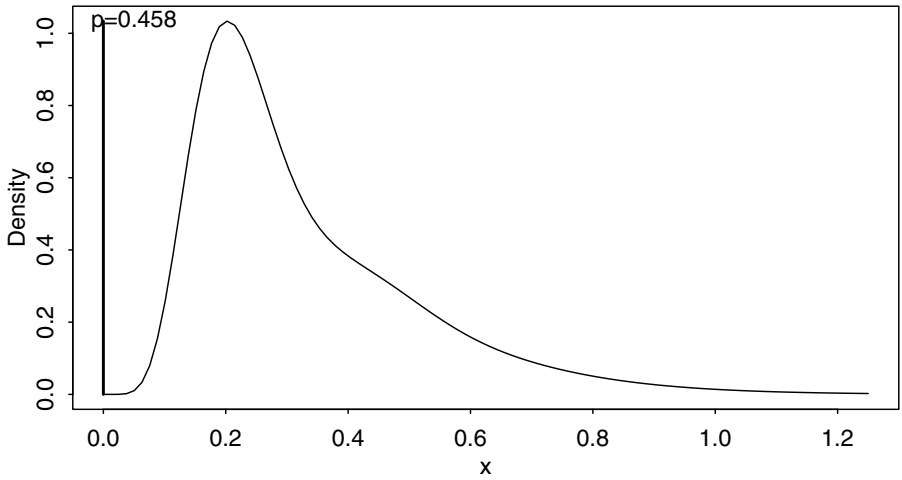


Figure 9.6 *Predictive density for the first zip code based on the fitted Tweedie GLM. Response is zero with probability 0.458 but otherwise takes a continuous positive value.*

Exercises

1. The relationship between corn yield (bushels per acre) and nitrogen (pounds per acre) fertilizer application was studied in Wisconsin. The data may be found in `cornnit`.
   (a) Make a plot of the data and comment on the relationship. Do you think the effect of nitrogen on yield is likely to be additive or multiplicative? Do you think yield will always increase with more nitrogen or will there be an optimum application of nitrogen?
   (b) Using a (Gaussian) linear model, represent the relationship between the yield response and the nitrogen predictor taking into account your answers to the previous questions. Check the residual vs. fitted plot and the QQ plot and comment on any anomalies.
   (c) Remove any outliers and repeat the model diagnostics.

(d) At what level of nitrogen does the maximum yield occur according to your model? Find a 95% confidence interval for the expected yield at this dose of nitrogen.

(e) Now develop a GLM for the data that does not (explicitly) transform the response. Pick a form for this model which is analogous to the Gaussian linear model just fitted. Check the same diagnostics as before.

(f) Remove any outliers and repeat the model diagnostics.

(g) Find the optimum level of nitrogen and form a 95% confidence interval for the expected yield.

(h) Make a plot of the data that shows the fitted curves from both models.

2. An experiment was conducted as part of an investigation to combat the effects of certain toxic agents. The survival time of rats depended on the type of poison used and the treatment applied. The data is found in `rats`.

(a) Make plots of the data and comment on differences between the treatments and poisons.

(b) Fit a linear model with an interaction between the two predictors. Use the Box-Cox method to determine an optimal transformation on the response. Can this optimal transformation be rounded to a more interpretable function?

(c) Refit the model using your chosen interpretable transformation. Check the standard diagnostics and report any problems.

(d) Is the interaction term significant? Simplify the model if justified and indicate which poison and treatment will result in the shortest survival time.

(e) Build an inverse Gaussian GLM for this data. Select an appropriate link function consistent with the previous modeling. Perform diagnostics to verify your choice. Which poison and treatment will result in the shortest survival time?

(f) Compare the predicted values on the original scale of the response for the two models. Do the two models achieve a similar fit?

3. Components are attached to an electronic circuit card assembly by a wave-soldering process. The soldering process involves baking and preheating the circuit card and then passing it through a solder wave by conveyor. Defects arise during the process. The data is found in `wavesolder`.

(a) Make boxplots of the number of defects for each of the predictors and comment on the relationships seen.

(b) Compute the mean and variance within each group of three replicates. Plot the variance against the mean. What is the relation?

(c) Fit a quasi-poisson model for the mean number of defects using the main effects of all the predictors. Which predictors seem to be significant and which not? Make a test to check whether the insignificant predictors can be removed from the model.

(d) Fit a gamma GLM with a log link for the variance response using all the main effects of the predictors. What problem arises?

(e) Now fit a standard linear model to the log of the variance. Check whether the same set of predictors can be removed from this model as previously.

(f) Try fitting the gamma GLM with a log link for the variances, now using a reducted set of predictors as determined from the previous question. Compare the fitted values from both models of the variance. Are they close?

(g) Examine the final models for the mean and variance. State the qualitative effect of the chosen predictors on the mean and variance.

(h) Compare the fitted values from the model for the mean with those for the variance. What is the relationship? Are the fitted variances consistent with what might be expected given a quasi-Poisson model?

4. Data were collected from 39 students in a University of Chicago MBA class and presented in `happy`. Happiness was measured on a 10 point scale.

(a) Plot the happiness response on each of the four predictors. Comment on the relationships.

(b) Rescale the response and fit an appropriate quasi-likelihood model. Which predictors are not statistically significant in your model?

(c) Consider dropping each of the predictors from the full model using an F-test to check the significance of each predictor. Remove the least significant predictor and repeat the consideration of single predictors relative to the current model. Iterate as necessary.

(d) Make a plot of the residuals and predicted values from your selected model. Interpret it. Which subject is the best example of someone happier than s/he deserves to be?

(e) Using your model, indicate how much familial income would compensate for a two level drop in `love`.

5. The `leafblotch` data shows the percentage leaf area affected by leaf blotch on 10 varieties of barley at nine different sites. The data comes from Wedderburn (1974).

(a) Make plots of the blotch response against the two predictors. Describe the relationships seen.

(b) Fit a binomial GLM with blotch as the response and site and variety as predictors. Find the deviance and degrees of freedom and comment on what this says about the fit of this model.

(c) Fit a quasi-binomial model and report the value of the dispersion parameter. Could this dispersion estimate be derived from the binomial GLM? Demonstrate how if possible.

(d) Plot the residuals against the predicted values. Interpret the plot.

(e) A better variance function is $\mu^2(1-\mu)^2$ and yet this is not one of the available choices in R. However, the effect may be obtained by the appropriate use of weighting. Define weights as a function of μ that, when used in conjunction with a variance function of $\mu(1-\mu)$, achieve the effect of a $\mu^2(1-\mu)^2$ variance function. Repeat the plot of the residuals and comment.

(f) Test the significance of each of the predictors relative to the full model using an F-test.

(g) There may be an interaction between sites and varieties although this is difficult to check. Which site and variety combination shows the greatest indication of an interaction?

6. Consider the `orings` data regarding space shuttle flights prior to the Challenger disaster from Chapter 3.

(a) Fit a quasi-likelihood model with an identity link to the data. You will need to rescale the response to $[0,1]$. You should use the usual binomial variation. Why does the model fitting fail?

(b) Now try fitting the same model but now setting the starting values for the regression coefficients using the `start` argument. You will need to make a valid choice for the coefficients. You may also find it helpful to set the argument `trace=TRUE` to check the convergence. What happens?

(c) Examine the summary output of your fitted model and determine the temperature above which the predicted probability of failure would be invalid.

(d) Fit a quasi-likelihood model but with the usual logit link. Compare this model to the identity link model. Determine which model is superior.

(e) Display curves showing the predicted probabilities as temperature varies for both models. Show these curves on a single plot, superimposed on the data.

(f) Compute the predicted probability of failure at 31°F for both models. Compare the results.

7. Fit the `orings` data with a binomial response and a logit link as in Chapter 3.

(a) Construct the appropriate test statistic for testing the effect of the temperature. State the appropriate null distribution and give the p-value.

(b) Generate data under the null distribution for the previous test. Use the `rbinom` function with the average proportion of damaged O-rings. Recompute the test statistic and compute the p-value.

(c) Repeat the process of the previous question 1000 times, saving the test statistic each time. Compare the empirical distribution of these simulated test statistics with the nominal null distribution stated in the first part of this question.

(d) Compare the critical values for a 5% level test computed using these two methods.

8. One hundred twenty-five fruitflies were divided randomly into five groups of 25 each. The response was the longevity of the fruitfly in days. One group was kept solitary, while another was kept individually with a virgin female each day. Another group was given eight virgin females per day. As an additional control the fourth and fifth groups were kept with one or eight pregnant females per day. Pregnant fruitflies will not mate. The thorax length of each male was measured as this was known to affect longevity. The data is presented in `fruitfly`.

(a) Make a plot of the data.

(b) Fit a standard linear model with longevity as the response and thorax length and activity as predictors, including an interaction term. Remove the interaction term if it is not significant.

(c) Plot the residuals and fitted values. Interpret the plot.

(d) Transform the response in an appropriate way and recheck the residual plot.

(e) Fit a gamma GLM with an appropriate link. Plot the residuals against the predicted values. Interpret the plot.

(f) Consider a fruit fly with mean thorax length in the isolated activity group. Use the fit from the model to plot the gamma density for the lifetime of such a fly.

9. The `truck` data concerns an experiment to optimize the production of leaf springs for trucks. A heat treatment is designed so that the free height of the spring should come as close to eight inches as possible. We can vary five factors at two levels each. A 2^{5-1} fractional factorial experiment with three replicates was carried out. The data comes from Pignatiello and Ramberg (1985).

(a) Plot the data and comment on the effect of the factors on the response.

(b) Fit a model with all two-way interactions for the height response. Why is it that not all two-way terms can be estimated?

(c) Use `step()` to choose a smaller model. What is the largest interaction term?

(d) Compute the predicted height for each of the 16 possible combinations of the factors. Order them in closeness to the target of 8 inches.

(e) Compute the variance in each group of three replicates. Use a gamma GLM with a log link to model the variances using all two-way interactions. Use `step()` to select a model. Comment on the factors which have an effect on the variance.

(f) Compute the predicted variance for each of the 16 combinations and display them in increasing order. Which combination produces the lowest variation?

(g) Discuss the problem of selecting a combination that comes close to the target of 8 inches while minimizing the variation.